



Carbon13

# Carbon13 Impact Report 2021/22

June 2024

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# Introduction

Carbon13 builds and backs ventures with a key focus on addressing the climate emergency. Entrepreneurship is a powerful creative force in the economy and one that can spur industrial transformation. This is exactly what we need to deliver against net zero at speed and with innovation that unlocks new technological and commercial trajectories.

We bring founders together in a programme designed to catapult climatetech into the global companies of tomorrow. Moreover, Carbon13 fosters a community by connecting startups with experts and like-minded individuals. This network enables knowledge sharing, innovation, and collaboration, creating a positive social ecosystem which can accelerate the development of climatetech solutions. With our origins in Cambridge, we were able to leverage our existing connections within a world leading tech cluster.

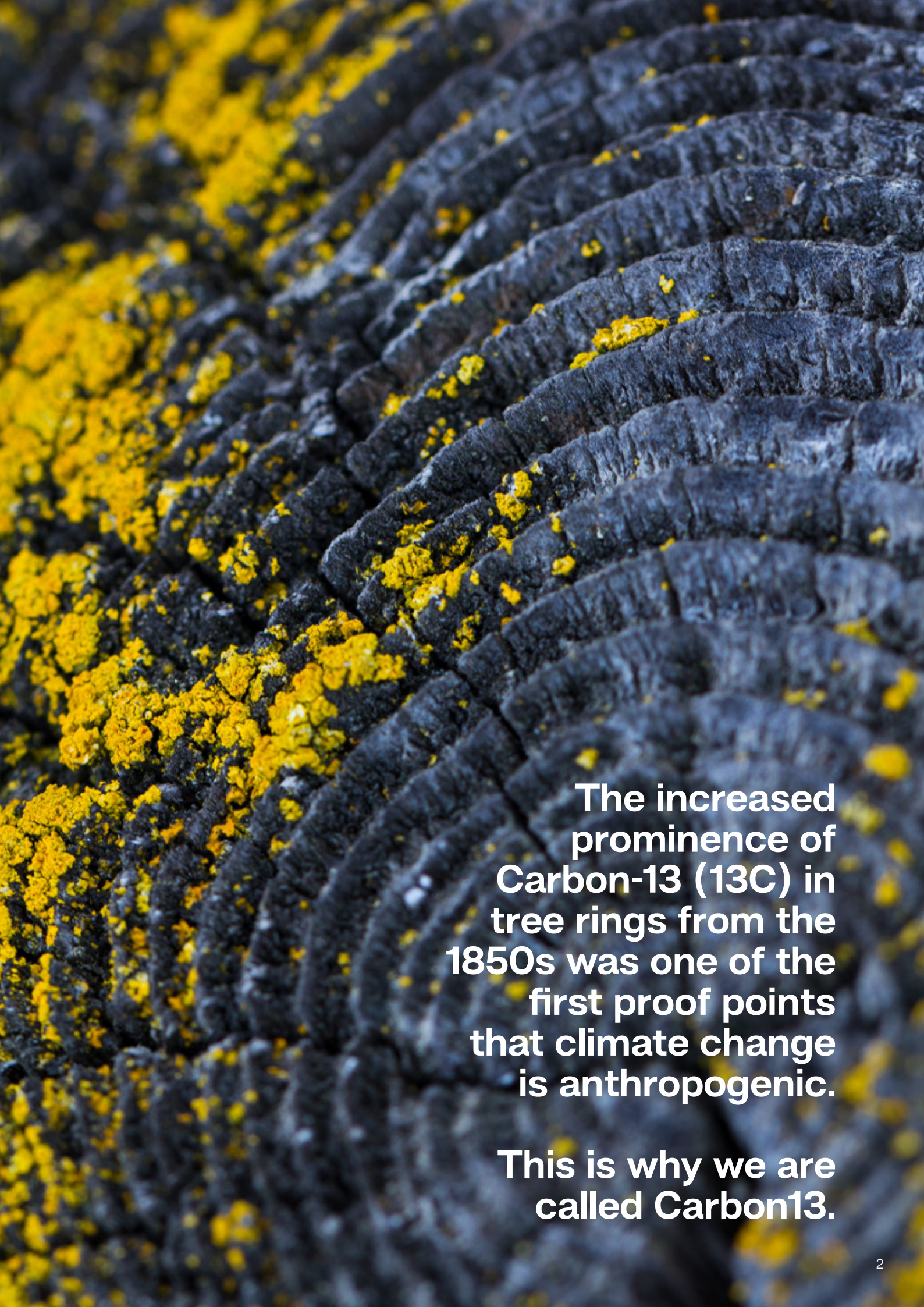
This report is for our activities in 2021 and 2022, when we launched and developed our programme, and founders were selected to build ventures capable of mitigating 10 million tonnes of carbon dioxide equivalents (MtCO<sub>2</sub>e) per year when at scale. We secured funds which we deployed into promising ventures created in the venture builder programme and built out a wide stakeholder group.

As a catalyst for problem solvers, Carbon13's impact on the environment, society, and the economy is substantial. While our primary impact remains focused on mitigating greenhouse gas emissions, we recognise the need to consider a broader range of impacts. We also acknowledge that how ventures are built matters. Founders have the opportunity to create the companies they wish existed and work inclusively with a broader range of perspectives which sparks new opportunities and ideas.

This is our first impact report and marks our commitment to sustainability and our mission to address climate change.







The increased prominence of Carbon-13 ( $^{13}\text{C}$ ) in tree rings from the 1850s was one of the first proof points that climate change is anthropogenic.

This is why we are called Carbon13.



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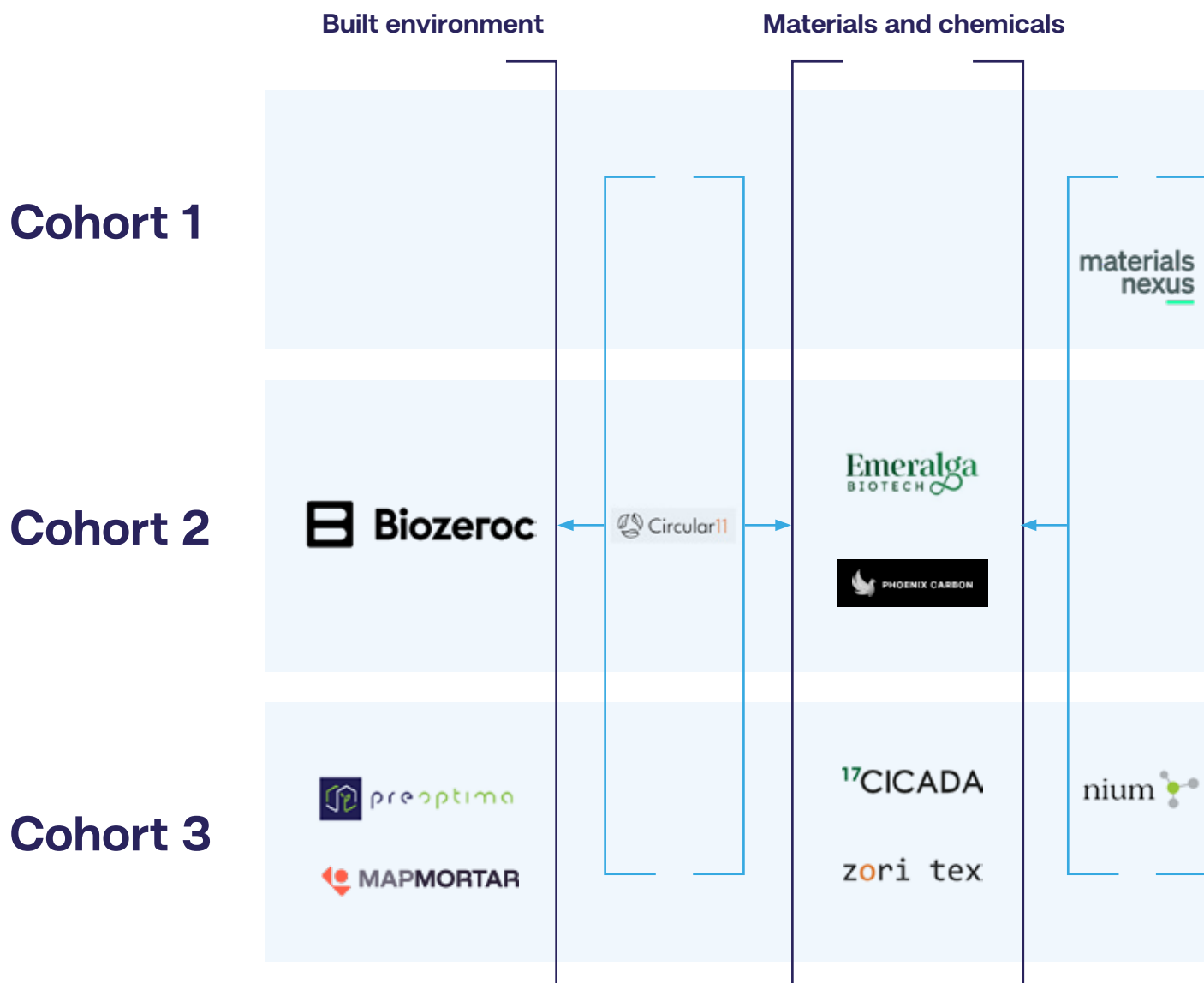
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# 01

## Our beginning

In 2019, we developed the concept of Carbon13. It all started with the Venture Builder Programme, which is delivered in three phases over eight months.

Phase 1 focuses on Teaming. The selected cohort participants have to find the ideal co-founder match to form a team capable of building a game-changing venture which can reduce carbon emissions by millions of tonnes. Only founders on the programme who successfully complete Phase 1 will continue to Phase 2: Validation. Phase 2 focuses on building proof points around the venture – especially its customers, the problem being solved, the value proposition and impact. At the end of Phase



2, teams meeting our criteria become eligible to present to the Investment Committee to decide which ventures receive investment and continue to Phase 3. Once in Phase 3, the focus is to help ventures build customer traction and momentum, including preparing a funding strategy for post-Carbon13 life and helping in the next round of finance.

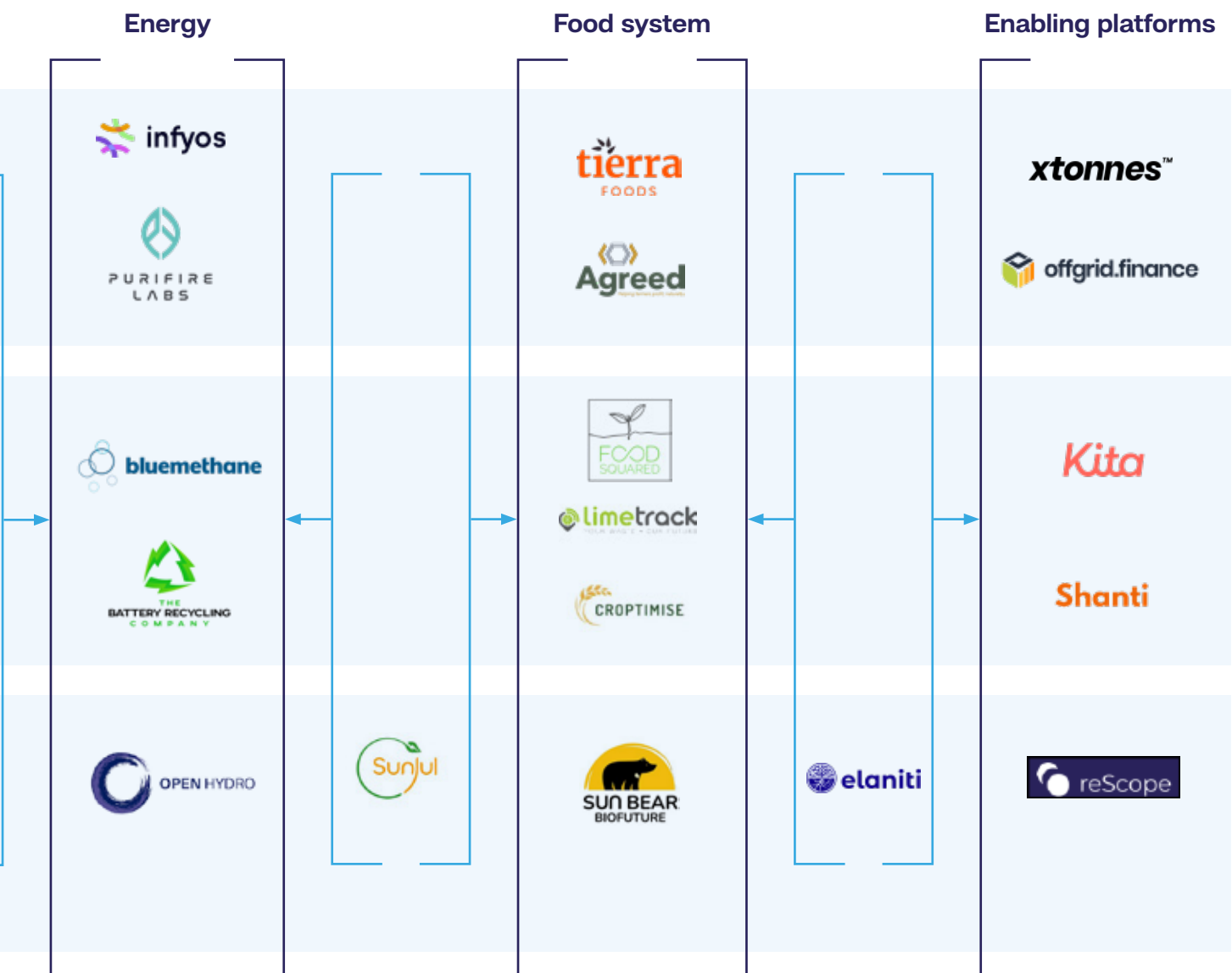
We started our first Venture Builder Programme in 2021 during the COVID pandemic in Cambridge, UK. This was a tough decision as we knew that teaming and

ideation are critical at the very beginning of our Venture Builder Programme and yet are some of the most challenging activities to do in a fully virtual environment. Nonetheless, with funding secured and having already delayed the start time due to COVID, we felt we, and the climate emergency, could not wait any longer. We came together at WeWork in Cambridge with our team of seven and worked together to make our venture builder for the climate emergency a reality. This first programme was an outlier, with some in-person activities when

permitted. With subsequent programmes, we have incorporated in-person events in Phase 1.

During 2021 and 2022 we made 32 investments in Cohorts 1 – 3.

This is Carbon13's first impact report covering 2021 and 2022. It is the start of our annual reporting process, which will progress as we develop more sophisticated and comprehensive ways of tracking impact in our own organisation and our primary source of Scope 3 emissions—our investments.



# 02

## Carbon13 and our statement of additionality

While we have seen huge progress with countries, companies and individuals making commitments to tackle climate change, we are not yet delivering action fast enough.

We also know that the type of transformation needed to achieve net zero requires disruptive and radical innovation.

In 2020, the International Energy Agency (IEA) quantified the need for technological innovation and investment for a cleaner and more resilient energy sector with net-zero emissions:

"Early-stage technologies play an outsized role. Around 35% of the cumulative CO<sub>2</sub> emissions reductions needed to shift to a sustainable path come from technologies currently at the prototype or demonstration phase.

A further 40% of the reductions rely on technologies not yet commercially deployed on a mass-market scale.<sup>1</sup> "Our thesis is that innovation is needed both to progress early stage technologies and to mainstream existing technologies in energy and other sectors. Some of this innovation will occur in products and processes, and some will occur in business models. We expect that a transformation to net zero will require radical and disruptive ideas that will occur on a scale that shifts industry and the economy.

Joseph Schumpeter reasoned that 'in general it is not the owner of stage-coaches who builds railways'<sup>2</sup> We also expect that net zero will prompt a new wave of entrepreneurs who are relatively unknown today but will scale their ventures to deliver globally recognised solutions.

## Our mission is climate

Our community is rooted in our mission to fight the climate emergency, and driven by our scalability ethos. This is where climate tech startups will succeed on a global scale.

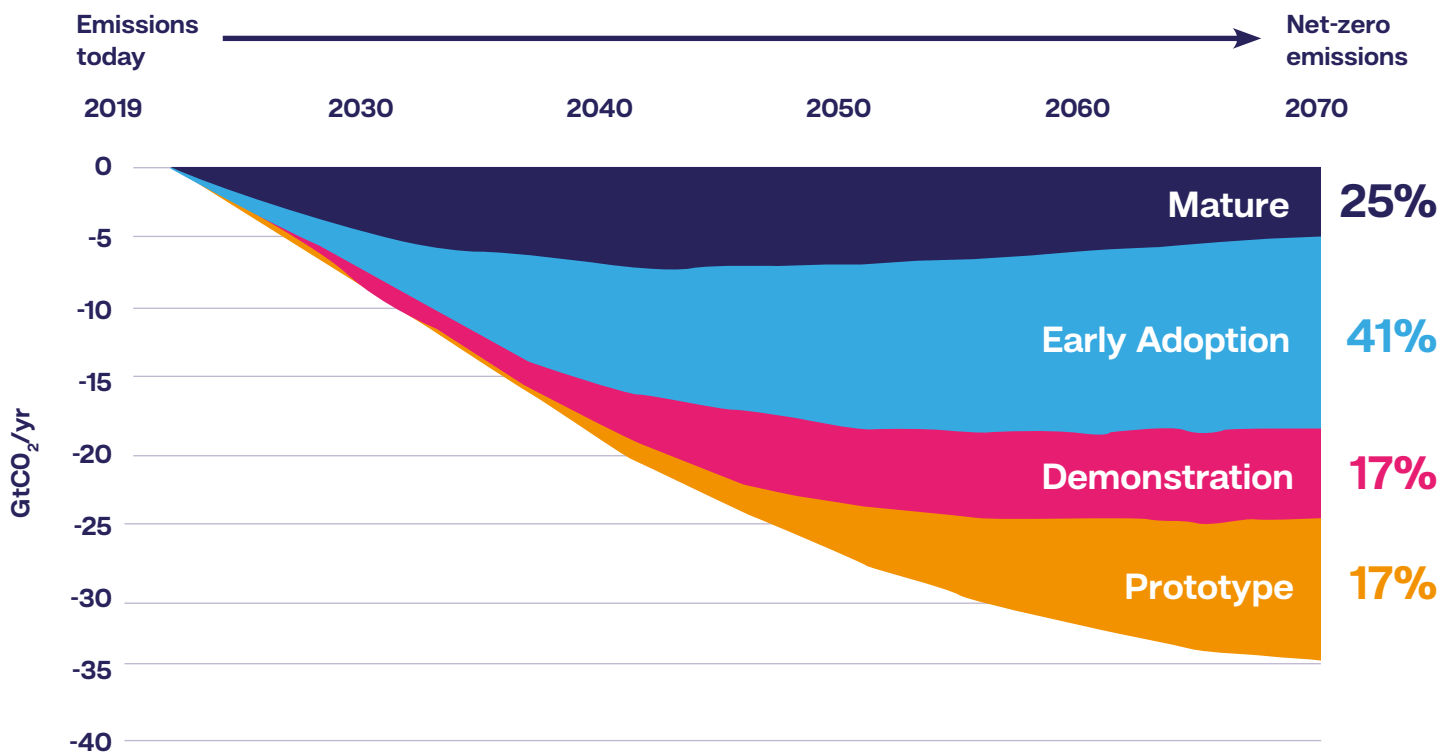


Figure 1 Global energy sector CO<sub>2</sub> emissions reductions by current technology readiness category in the Sustainable Development Scenario relative to the Stated Policies Scenario (p.28, IEA 2020) Notes: Percentages refer to cumulative emissions reductions by 2070 between the Sustainable Development Scenario and the Stated Policies Scenario enabled by technologies at a given level of maturity.



# 02a

## Additionality

To deliver our mission we need to understand our own additionality<sup>3</sup> which is essentially our theory of how we create change through our activities.

**There are four main areas where Carbon13 creates additionality:**



"When I turn up at Carbon13, there's this whole slew of CVs and I was like 'jeez, am I the only one without a PhD?' and then I realised, it's probably a good thing because if we all had PhDs, we'd all be doing the same thing or bringing the same things to the table."

**Frankie Fox,  
Founder FoodSquared**

### **1. Creating ventures that otherwise would not exist (relevant to our Venture Builder Programmes)**

**Bringing people together:**  
Carbon13 connects founders that otherwise would not have met, which sparks the creation of new ventures.

**Entrepreneurial outreach**  
Carbon13 engages potential founders who were unsure about becoming entrepreneurs to join the force to fight the climate emergency through entrepreneurship.

### **Enabling founders to find a team**

We connect founders who wanted to team prior to building a venture, to find their ideal cofounder match.



### **2. Supporting ventures to be more impactful**

#### **Accelerating growth and scale**

- Developing a scalable plan with input from experienced people from our support network who have done it before.
- Getting the right input at the right time.
- Facilitate access to finance
- Support team effectiveness and diversity (more diverse teams perform better!).

#### **Embedding impact**

- We train every founder to incorporate impact since the ideation phase and manage and communicate impact. This is because we believe this is the only way to embed impact. Every founder should be a catalyst for change.



<sup>3</sup> Additionality is the extent to which something happens as a result of an intervention that would not have occurred in the absence of the intervention.



### 3. Creating new knowledge which benefits founders who do not progress with a venture

We have anecdotal evidence that founders who have not progressed in Carbon13 still value the experience and can re-engage with work in a more impactful way. One ex-founder re-joined her previous company and spoke about being able to engage with startups and support them, something she had not done prior to Carbon13.

### 4. Address the lack of diversity in entrepreneurship

- Active outreach on marketing to under-represented groups
- Programmatic design to be more inclusive towards varied founder types
- Active support to female founders

There could be counter arguments to these points of additionality. For example, our original venture builder programmes included some early stage pre-formed teams (two or three in a pool of 20 to 25 teams). Arguably, these would have become ventures with or without Carbon13. However, the majority of ventures in our venture builder programmes were created by teams that were unknown to each other before Carbon13.



**"As a scientist on the Venture Builder, yes it was great to join a group which also had lots of other scientists and engineers, but I was also taken out of my comfort zone and challenged by amazing entrepreneurs and commercial founders. That's what took it from an idea to a business."**

**Dr Yubiao Niu,  
Co-Founder & CTO @Nium**



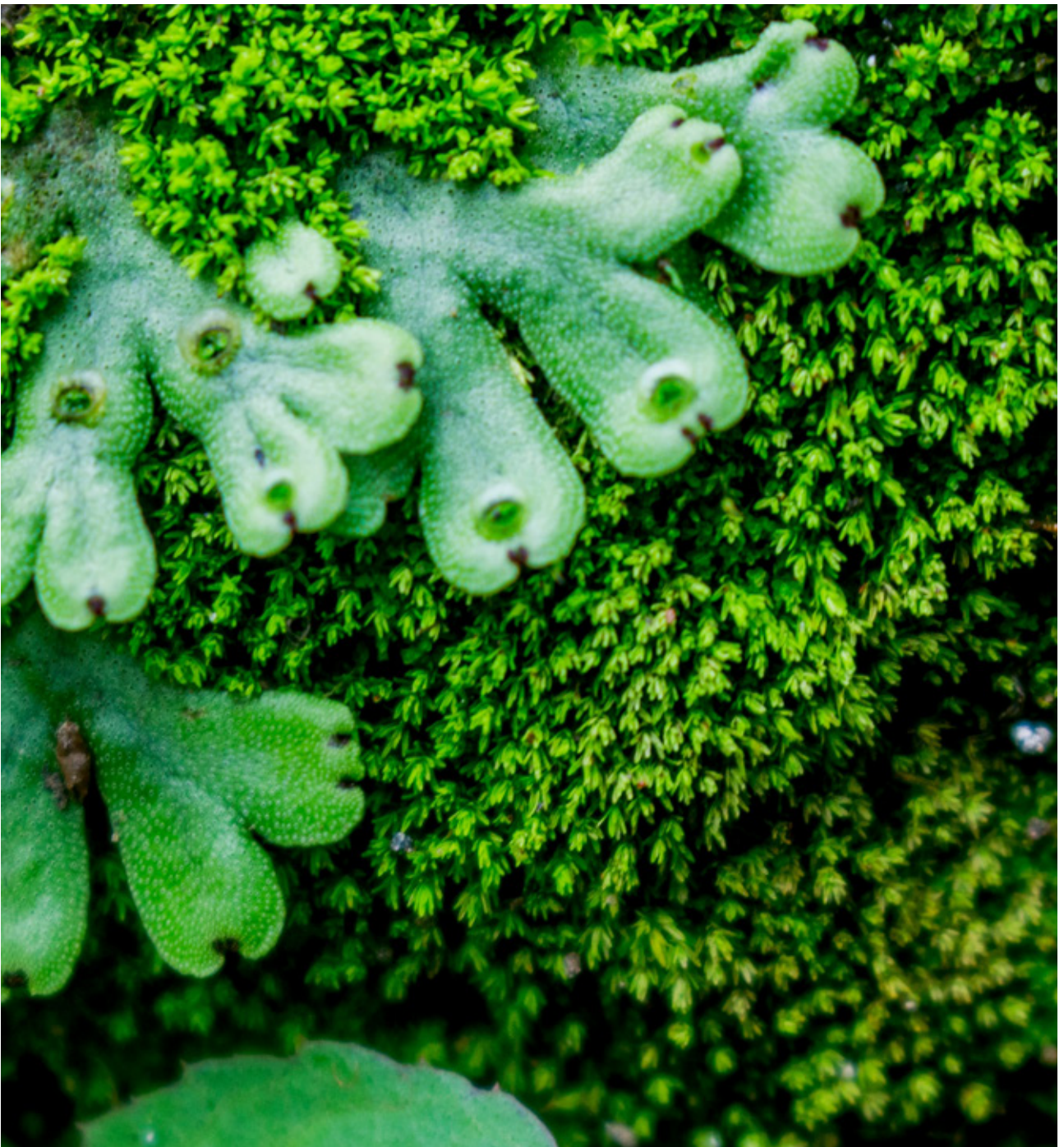
**"Joining Carbon13 was a breath of fresh air as it enabled me to execute much faster than any other time in my career. One of the biggest parts of a company is the team and finding others with a similar vision has transformed my business from idea into reality."**

**Jonathan Bean,  
Founder and CEO of  
Materials Nexus**



# 02b

Our challenge to mitigate  
10 million tonnes of carbon  
dioxide equivalents





**Dr Nicky Dee introduces the 10MtCO<sub>2e</sub> to founders of the Venture Builder UK programme**

We ask every venture that goes through a Carbon13 programme to work towards mitigating net 10 million tonnes of carbon dioxide equivalents (10MtCO<sub>2e</sub>) per year when at scale typically within 7-12 years of incorporating.

To put this differently, this means mitigating 2% emissions from a market or sector emitting 0.5GtCO<sub>2e</sub>.

We focus on 10MtCO<sub>2e</sub> rather than the mitigation potential as a hurdle rate for our ventures as we know there are hard to abate areas where a venture with a compelling and defensible position will be able to secure an unfair advantage and a higher percentage of the market.

Working towards a plan to mitigate 10MtCO<sub>2e</sub> demands a growth mindset. Developing a scaling mentality from day one is essential to address an urgent and sizable problem.

This isn't just about tackling greenhouse gases for the sake of impact. Commercial drivers are emerging as a result of the cost of inaction on climate change being more than the cost of action. This commercial opportunity will crystallise from the technical, business model, and regulatory experiments currently being taken. Our hypothesis at Carbon13 is that tackling greenhouse gas emissions is good business, even if we don't yet understand which business models and technologies will win out. Engaging with a climatetech community on the forefront should help give an edge. However, we believe that the only way to create an impact and a commercial proposition is to train entrepreneurs in both.



# 02c

## Training every founder

A core element of Carbon13 is training every founder to understand and communicate their impact potential.

One of the unique things that we are doing at Carbon13 is we are not expecting our founders to outsource this knowledge, we are very much training all of our founders, so they have access to the latest tools and techniques, so that they can assess their carbon mitigation potential, understand how to communicate it and of course be transparent in how they are measuring it.

As custodians of their businesses, founders are best placed to drive impact. This isn't isolated to just understanding greenhouse gas emissions and related impacts across the Sustainable Development Goals (SDGs), but also how a carbon proposition is tied to a commercial proposition so it is embedded in the core business.

The rationale has a corollary to the need for every founder to understand their finances and business model so they can remain alert to internal and external forces that prompt the need to shift or pivot the business.



# 02d

## Building and Pitching a Carbon Case

Since starting Carbon13, we have amended how we tackle the evaluation of impact within our ventures. While we have remained consistent with the need to train our founders, we have experimented with tools and approaches that our entrepreneurs can use.

Cohort 1 and 2 were required to incorporate their case for impact within the Investment Case. To support this, we also provided signposting to tools in addition to paying for access to a Life Cycle Assessment tool designed for non-experts.

As a result of the experiences using these tools and approaches, which didn't quite fulfil the needs and circumstances of our ventures, we developed our Carbon Case as a parallel requirement that each of our ventures had to submit alongside their Investment Case, which was introduced in Cohort 3 of our Venture Builder Programme. While the name of this document infers a focus on greenhouse gas emissions, the Carbon Case prompts a much broader view of impact as we recognise the complexity and interconnectivity of a range of impacts. Reference points include Planetary Boundaries and the SDGs.



# 03

## Emissions boundaries at Carbon13

A venture builder programme typically takes around eight months. The different cohorts and phases are mapped below against our 2021/2022 footprint.

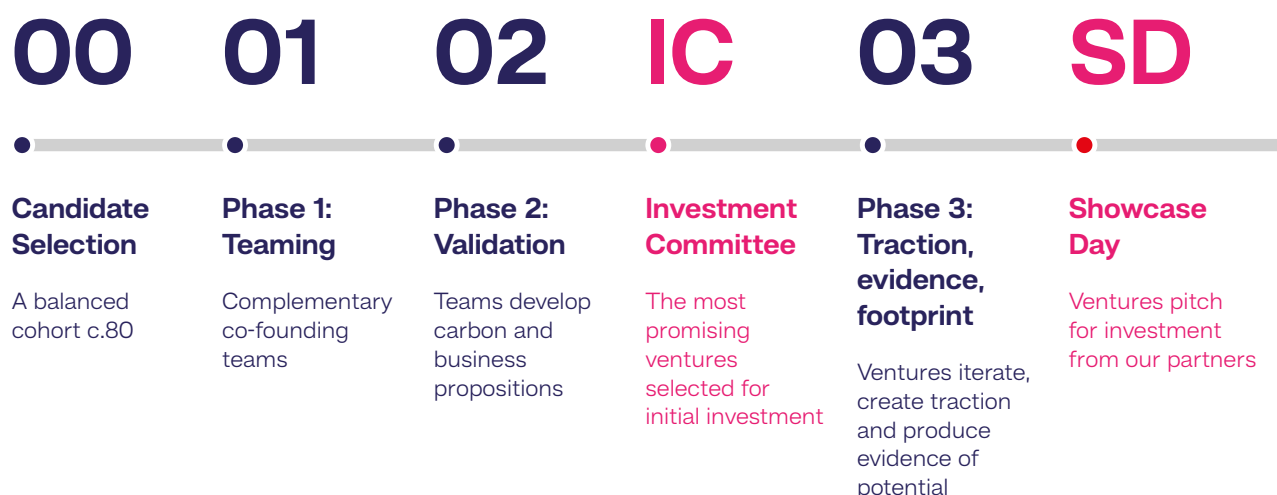
Phase 1 is focused on teaming and some climatetech building blocks, with individuals progressing to Phase 2 if they form a venture team. In Phase 2, teams work to develop their business concept and validate their ideas through customer discovery, innovation development and roadmap, and demonstrating team effectiveness.

A critical element is building out their Carbon Case in conjunction with their Investment Case. At the end of Phase 2, the teams pitch for pre-seed investment from Carbon13 and access to Phase 3 of the programme.

Since we start with individual founders and end with ventures the way we attribute emissions to Scopes changes throughout the programme. In Phase 1, participant travel to our events is part of our Scope 3 emissions, after which founders account for their travel emissions as part of their venture's Scope 3 emissions.

# The Venture Builder Programme

**Candidate Applications** c.600 candidates



## 2021

|                    |                                                      |
|--------------------|------------------------------------------------------|
| Cohort 1 - Phase 1 | 16 <sup>th</sup> Mar - 30 <sup>th</sup> Apr          |
| Cohort 1 - Phase 2 | 7 <sup>th</sup> May - 29 <sup>th</sup> Jul           |
| Cohort 1 - Phase 3 | 2 <sup>nd</sup> Aug - 1 <sup>st</sup> Dec            |
| Cohort 2 - Phase 1 | 20 <sup>th</sup> Sep - 29 <sup>th</sup> Oct          |
| Cohort 2 - Phase 2 | 1 <sup>st</sup> Nov 2021 - 27 <sup>th</sup> Jan 2022 |

## 2022

|                    |                                                       |
|--------------------|-------------------------------------------------------|
| Cohort 2 - Phase 3 | 31 <sup>st</sup> Jan - 31 <sup>st</sup> May           |
| Cohort 3 - Phase 1 | 26 <sup>th</sup> Apr - 3 <sup>rd</sup> Jun            |
| Cohort 3 - Phase 2 | 6 <sup>th</sup> Jun - 29 <sup>th</sup> Jul            |
| Cohort 3 - Phase 3 | 9 <sup>th</sup> Aug - 9 <sup>th</sup> Dec             |
| Cohort 4 - Phase 1 | 19 <sup>th</sup> Sep - 28 <sup>th</sup> Oct           |
| Cohort 4 - Phase 2 | 31 <sup>st</sup> Oct 2022 - 20 <sup>th</sup> Jan 2023 |



# 03a

## Carbon13 Scope 1 & 2 Emissions

Carbon13 follows guidance from the Green House Gas Protocol (GHGP) but also strives to take a forward looking position towards our emissions.

For example, in 2021, we leased an office at WeWork (Cambridge), later moving to Eagle Labs. Carbon13 does not own these workspaces, and while we have control over lights, sockets, kettle, etc., we do not have ownership and, therefore, lack direct control over electricity sourcing, fugitive emissions, and efficiency. This could mean that the Scope 1 and 2 emissions associated with this space can be considered Scope 3, but we have allocated them to Scope 1 and 2.

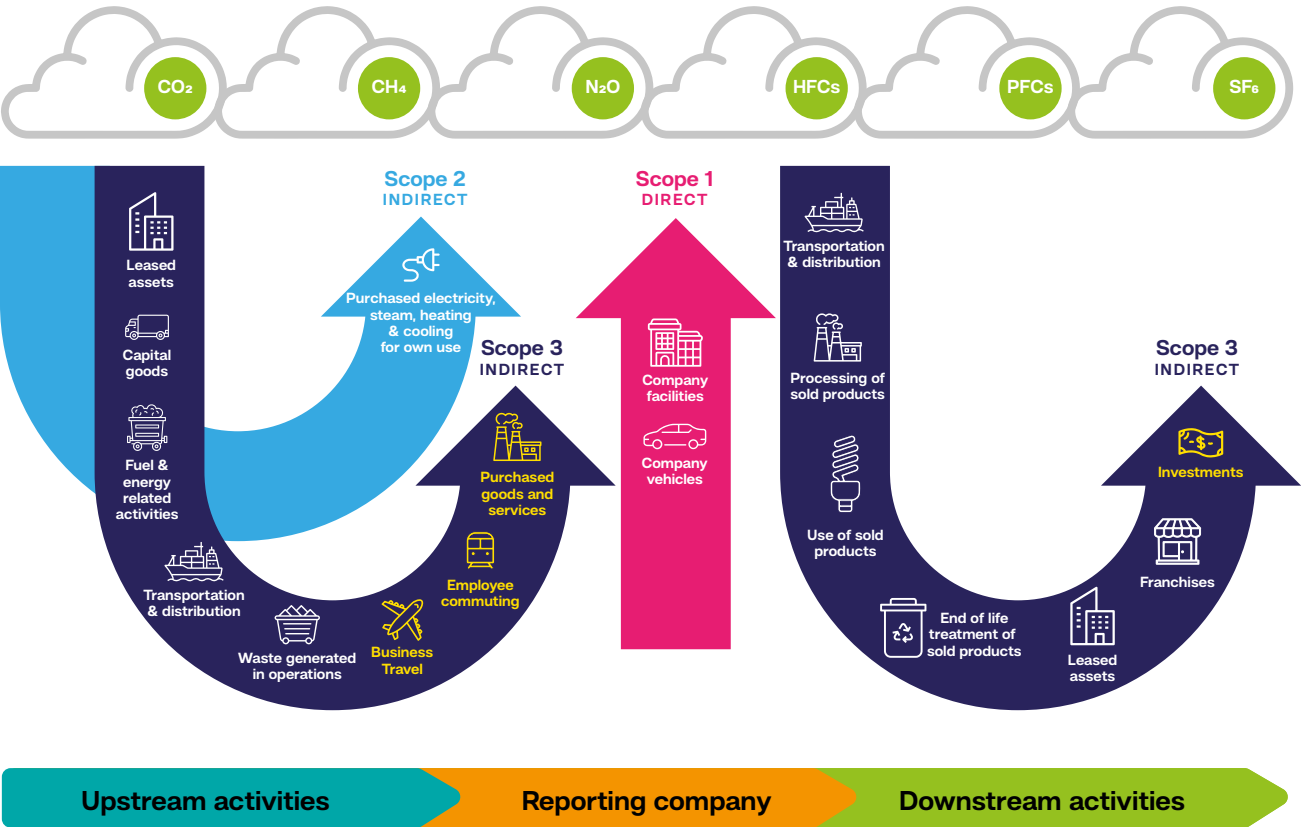
In alignment with GHGP, we have used standard grid emission factors. Barclays has a power purchase agreement for 250GWh of Renewable Energy Guarantees of Origin (REGOs) certificates<sup>4</sup>, which is allocated to the Eagle Labs and other facilities.

There is a reduction in Scope 1 and 2 emissions in 2022 versus 2021 as a result of taking on a smaller office space and moving to a remote first approach with co-working opportunities and company activities. Remote working is moved into Scope 3 but also represents a reduction in per capita emissions versus office based working, which is the principle cause for reduced emissions despite an increase in the number of employees.

<sup>4</sup> REGO certificates are administered by Ofgem with a goal to provide greater transparency on the percentage of power generated from renewables. REGOs are being continuously reviewed for their ability to drive additional renewables. For more details on Barclays see here - <https://home.barclays/news/press-releases/2022/02/barclays-signs-10-year-power-purchase-agreement-for-uk-operation/>

Overview of GHG protocol scopes and emissions across the value chain

Scope 3 has the most relevant items for Carbon13 in yellow



Adapted for Carbon 13 from original source ghgprotocol.org

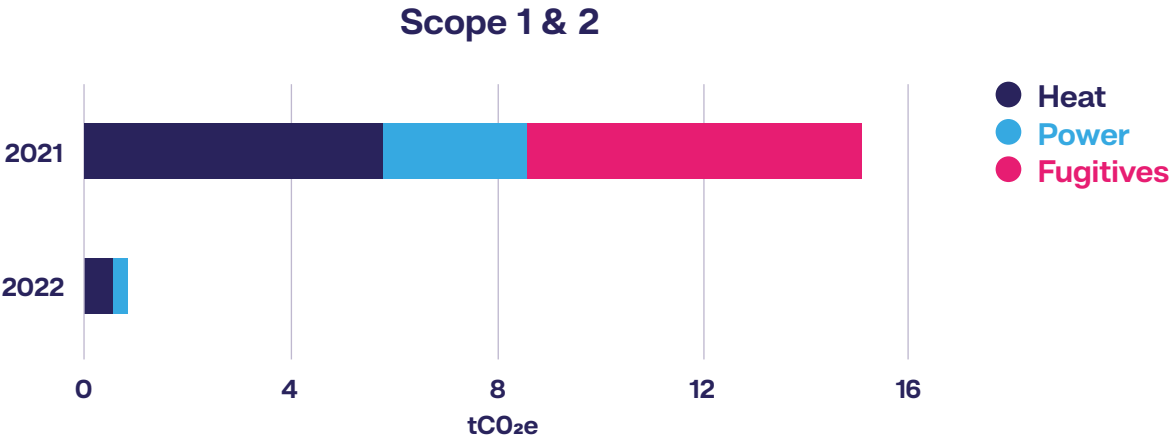


Figure 2 Carbon13 Scope 1 & 2 emission for 2021 & 2022. Middle figures are used. For full range of figures see the appendix



# 03b

## Scope 3

To gather and synthesise our emissions, we use xtonnes<sup>5</sup>, a carbon action software developed in Cohort 1

xtonnes uses data research, science and sustainability expertise to assess an organisation's carbon footprint.

Assets and activities data provided by the organisation are assessed using carbon factors from government sources, reputable third party research, and xtonnes' proprietary modelling, with the most relevant factors being applied to our business.

Where proxy data exists for how much an asset or activity has been used or carried out, this creates a "modelled" footprint, giving the most likely, lower and upper case possibilities. Where data exists that measures the usage of an asset or completion of an activity, this creates a measured footprint, which typically receives a maximum quality score, and produces a single figure as opposed to a range.

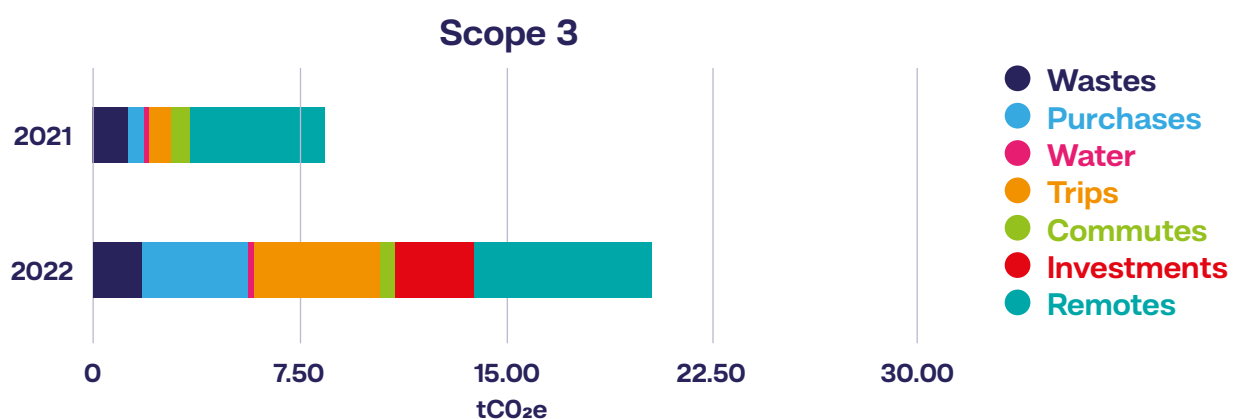


Figure 3 Carbon13 Scope 3 emissions for 2021 & 2022. Middle figures are used. For full range see the appendix.

### Key:

**Wastes:** Emissions associated with the disposal of waste generated by the organisation.

**Purchases:** Emissions associated with the production, transportation, and disposal of goods and services purchased by the organisation.

**Water:** Emissions associated with the organisation's water consumption and wastewater treatment.

**Trips:** Emissions associated with employee travel for business purposes.

**Commutes:** Emissions associated with employees commuting to and from work.

**Investments:** Emissions associated with the organisation's investments.

**Remotes:** Emissions associated with remote operations that are carried out or situated away from the organization's main location.

<sup>5</sup> <https://carbonthirteen.com/our-portfolio/platforms/xtonnes/>

# 03c

## Carbon13 employees and our programmes

### Employees

A significant difference between 2021 and 2022 is derived from the difference in how the emissions per employee are calculated.

**Primarily the figures reflect an increase in home-working and the lower emissions which are associated with this according to new guidelines:**

- Every employee working at home is assumed to use energy only for laptop/PC, monitor, phone, printer, and (a small amount of) lighting.
- Home heating emissions are only considered on an 'incremental' basis. It is assumed that one-third of employees have at least one household member who would normally remain at home during the day.

As a result of these assumptions, home energy consumption is expected to be lower than office consumption. The logic is that only two-thirds of employees working from home lead to heating-related emissions that wouldn't have already taken place (ie: incrementally). Combined with the assumption that no air-conditioning takes place at home, this likely leads to a much smaller per employee carbon.

<sup>6</sup> Office emissions include higher energy demands for air conditioning, increased number of electrical appliances and lighting of communal and meeting spaces

Founders on the Venture Builder programme in Jesus College, Cambridge



## Programmes

We did a baseline travel survey of one of our cohorts to assess the average travel distribution and frequency from the cohort in Phase 1 of the programme.

**Phase 1 is essential for all founders and includes three blocks of three in-person days in Cambridge depending on the programme.**

We also have some events within our programme that fall under Scope 3 emissions. The most significant of these are probably our two Showcase Days, one in Cambridge and one in London. While the nature of these events continues to evolve, the Cambridge showcase attracts the most participants and is associated with the most significant emissions.

## Venture Builder

|         |                                                       |
|---------|-------------------------------------------------------|
| Phase 1 | 3 X 3 day in-person events                            |
| Phase 2 | Garden Party<br>Optional in-person investment pitches |
| Phase 3 | Showcase Days                                         |



# 03d

## Investments

For 2021 and 2022, we are including investments from Cohort 1-3, totalling 32 invested ventures.

Once ventures leave the programme, we reach out to them as part of our portfolio management. This impact report includes data from Cambridge Cohorts 1 and 2 in 2022.

**Carbon13 is often the first-cheque investment, which means we are backing ventures at the very beginning of their journey. We assess impact as follows:**

- Current impact – which is collected through gathering data on the companies' Scope 1, 2, and 3 emissions.
- Planned impact – each of the ventures in 2021 & 2022 put together a proposition to achieve 10MtCO<sub>2e</sub> typically within 7 – 12 years after launch.

For current impact, the GHG Protocol Accounting and Reporting Standard provides unclear guidance on when an investment should be considered material. Since building, supporting and investing in climate ventures is Carbon13's core business, we feel it is critical that we include these emissions. Carbon13 Limited has options in the companies in which we invest in addition to the equity held by our investments. We make the assumption that our options will convert and, therefore, include our potential share position within our Scope 3 emissions.

We are currently developing our portfolio management system, including tracking investment and impact criteria. We expect to provide an update later this year.

### Planned Impact

There are two crude measures of whether our ventures are on track to mitigate 10MtCO<sub>2e</sub> per year at scale:

- Are they still operational?
- Have they secured follow on funding?

Within Cohorts 1 – 3, as of December 2023, we knew c.90% are still operational and c.60% have secured follow on funding. We expect impact will follow a similar trajectory to revenue, where revenue can be non-existent for pre-seed ventures as they build their product or process to underpin future growth. For this reason, the planned impact will significantly exceed the current impact for all ventures currently in our portfolio (Fig. 5). Planned impact according to the ventures that have secured follow on funding, is estimated at c. 180 MtCO<sub>2e</sub> for Cohorts 1 – 3.



**Michael Langguth,**  
**Co-Founder at Carbon13**

at the Climate Innovation Showcase  
for Cohort 3 of the Venture Builder  
UK at the Cambridge Union

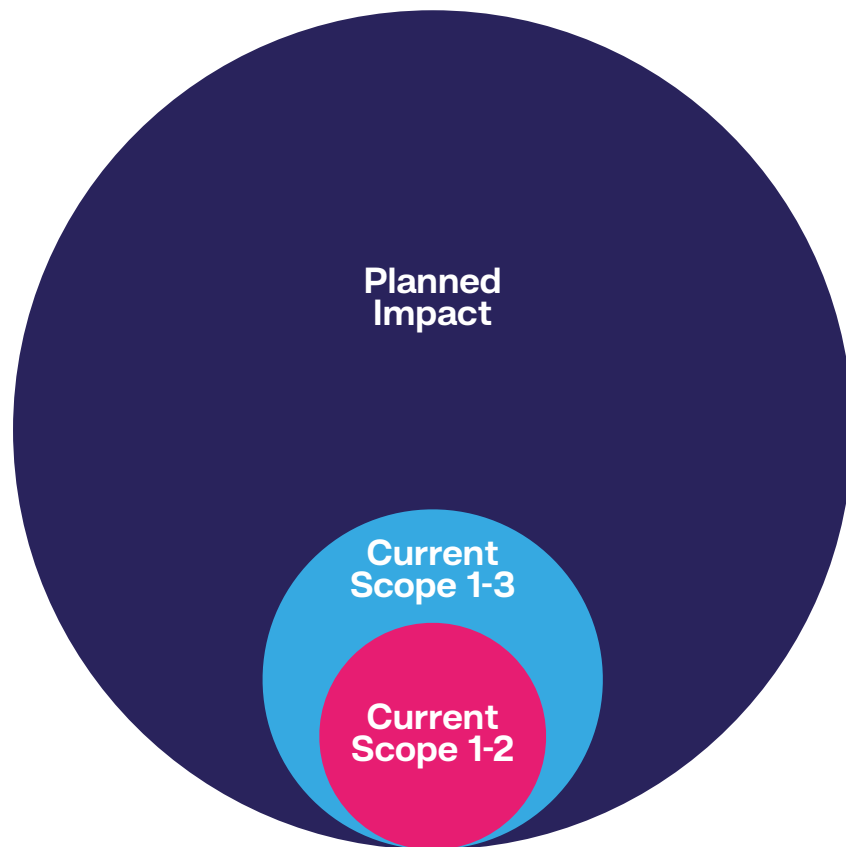


Figure 5 Illustration of planned vs current impact (not to scale)

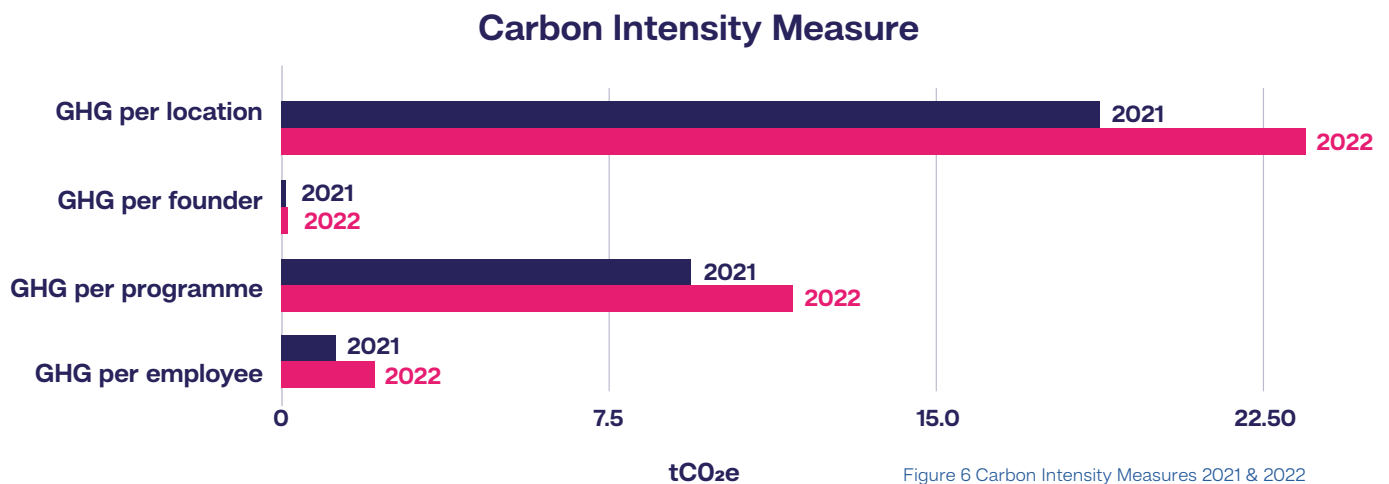


Figure 6 Carbon Intensity Measures 2021 & 2022

## Carbon Intensity

As a growing organisation, we need to review our organisational boundaries and what this means for our target setting. We do need to use carbon intensity figures but are mindful of

the over-reliance on carbon intensity instead of focusing on absolute reduction in emissions. For this reason, we will not use a single carbon intensity measure but use a basket of figures, which we will continually review.

For example, as we anticipate opening new locations, we expect a per-location figure to help provide learning opportunities on decarbonisation.

# 04

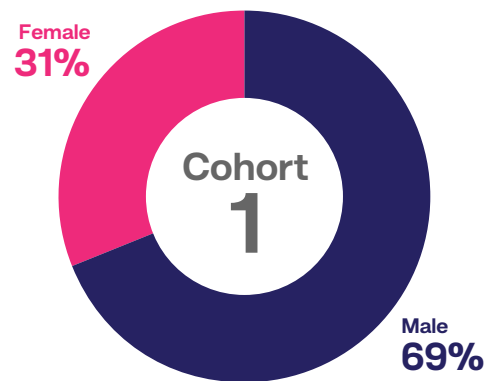
## Diversity in entrepreneurship

Carbon13 welcomes and values diversity in its staff, programme participants, mentors, investors, board members, board advisors, sponsors, and partners and seeks to create an atmosphere that is inclusive and respectful of differences, and is fair.

We do this because, in addition to its being the right thing to do, it also contributes to high-performing teams who can deliver valuable insights to achieve our mission to remove or reduce greenhouse gas emissions at scale and fast.

### Summary of Principles

Carbon13 is committed to equality of opportunity and fair treatment both as an employer and as a provider of services. We also work to understand the contribution we can make to equality and to develop creative ways we can support it.



### Carbon13 will:

- Ensure that policies and processes reflect Carbon13's commitment to equality and its aspiration to equality.
- Support an ethos of fairness, courtesy and respect.
- Ensure that everyone is treated fairly regardless of their race, colour, nationality, ethnic or national origin, sex, family responsibilities, disability, age, sexual orientation, gender reassignment, marriage or civil partnership, pregnancy or maternity, political or religious beliefs, socio-economic group, trade union activity or background.
- Ensure that members of staff are trained and promoted on the basis of their abilities and the requirements of the job and are employed and retained in a non-discriminatory manner.
- Monitor and review regularly the operation of this policy. Carbon13 will aim to monitor all work to ensure fair treatment and identification of patterns of under-representation. Every reasonable effort will be made to develop appropriate strategies to address patterns of under-representation. Carbon13

<sup>7</sup> The\_impact\_of\_business\_accelerators\_and\_incubators\_in\_the\_UK.pdf (publishing.service.gov.uk)

<sup>8</sup> clinton.pdf (un.org)

<sup>9</sup> <https://www.kauffmanfellows.org/journal/female-founders-hire-more-women>

<sup>10</sup> World Economic Forum Annual Meeting in Davos 2024



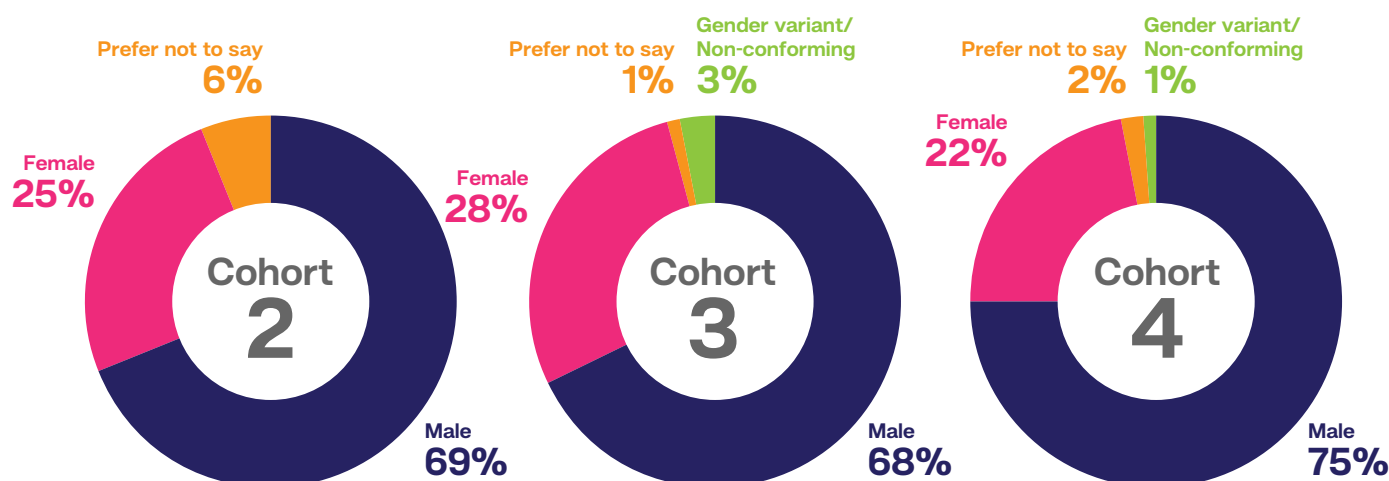


Figure 7 Founder gender split on the cohort

wishes to provide a harassment-free experience for everyone, regardless of gender, gender identity and expression, sexual orientation, disability, physical appearance, body size, race, group, age or religion. We will not tolerate harassment in any form.

### Tackling the Gender Gap: Elevating Gender Diversity as Our Primary Focus

As all aspects of Diversity, Equity, and Inclusion (DEI) are fundamental pillars of our organisation's ethos, we acknowledge their web of intersectionality and understand the complexity inherent in fostering an inclusive space.

However, given our limited resources as a small organisation, we have strategically prioritised initiatives aimed at enhancing gender diversity. We recognise that mentoring, resources, networking, and funding opportunities that venture builders and accelerators offer can help address the equity gap.<sup>7</sup> Climate startups encounter specific hurdles, like long-term return on investment

and regulatory shifts, which accelerators can assist female entrepreneurs in navigating with valuable guidance and support.

This segment of the impact report, thus, while underlining our commitment to a broad spectrum of social concerns, particularly emphasises our endeavours to reduce gender disparities in startup cofounding teams.

This approach is based on the understanding that addressing gender inequality has a cascading effect, positively influencing various other social aspects, from economic participation to increasing innovative solutions for problems that impact us all. Investing more in female entrepreneurs has the potential to yield significant societal benefits, as research shows that women often allocate a higher portion of their income towards their families compared to men.<sup>8</sup> Female-founded businesses could also provide more job opportunities to women once startups with at least one female founder hire 2.5 times more women than male-founded businesses.<sup>9</sup>

However, we are aware that our commitment to DEI extends beyond gender alone. As we progress, we remain dedicated to expanding our efforts to encompass broader diversity dimensions for a more comprehensive and equitable future.

Despite proof that organisations with greater gender diversity at the governance and leadership level outperform the ones with less, female representation in startup founding teams ranges from around 10% to 20%. In 2022, women-founded companies received just 2% of VC investment, while only 15% of VC's 'cheque-writers' were women.<sup>10</sup> Efforts to reshape this reality and enhance female representation in our programmes, and consequently in the startup world, remains a core focus. Across Cohorts 1 to 4, around 25% of the participants were female, marking a positive step forward. However, we acknowledge the journey toward gender parity is extensive, and we have much more to do to reach a female critical mass in our cohorts.

## 04 Diversity in entrepreneurship



**Dr. Nimisha, CSO of Elaniti,  
with co-founder Utkarsh Vaidya  
from Cohort 3**



**Tony To and Sarah Montgomery,  
co-founder and CEO of Infyos  
from Cohort 1**



**Kelly Price, co-founder and CEO  
of Agreed Earth from Cohort 1**

While investment decisions are made according to strict criteria, it is interesting to note that around 60% of Carbon13's invested companies have women in the founding teams, and 65% of those have female CEOs.

Our efforts to have more female representation, however, have to go beyond the candidates' selection process. Through targeted initiatives, programme sessions, and partnerships, we have concentrated our resources and energy on advancing the cause of gender equality in all its dimensions.

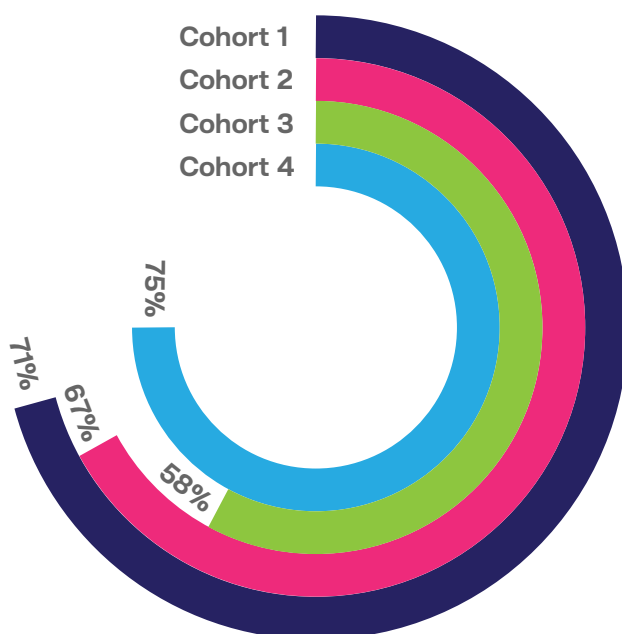


Figure 8 Percentage of founding teams with women

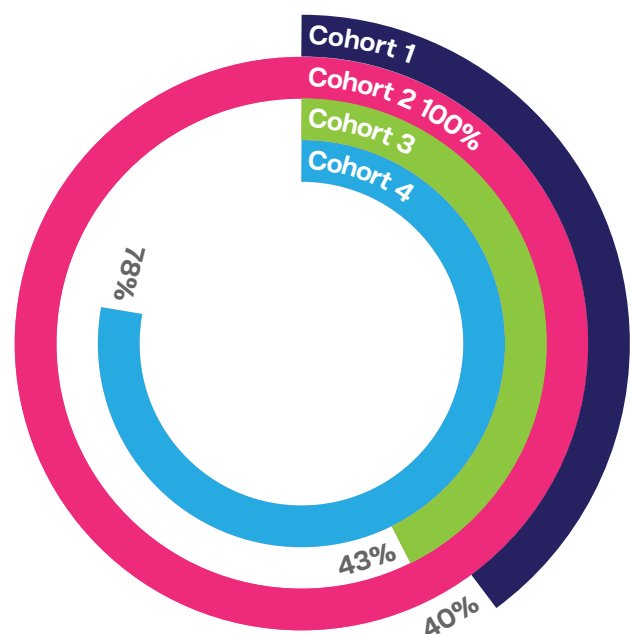


Figure 9 Percentage of invested teams with female CEOs

## 04 Diversity in entrepreneurship

Currently, the fight to reach Net Zero is missing out on some of the most desperately needed solutions because this industry is excluding those who will be most affected by climate change, therefore, who should be involved in building them.<sup>11</sup> This is why we're working to keep inspiring, supporting, training, and funding female founders and other under-represented groups.

To make sure the much-needed support for female founders is taking place, Carbon13, as an organisation, also has an important role to play behind the curtains. In 2021, 73% of the company's employees were female. And 50% of the senior leadership roles were occupied by women. Furthermore, 25% of our Entrepreneurs in Residence (EiRs), 40% of our board, and 15% of our advisory board were female. In 2022, 54% of the company's staff were female, while 25% of the senior leadership were women. Female representation in our EiRs was 40%, 20% on our board, and 38% on our advisory board. The gender split of our Domain Experts, will be made available in the next report.

### **EDI beyond gender disparities: The importance of having different Ethnicities**

In Carbon13, besides our efforts to fight gender disparities in the climatetech world, we recognise that diverse ethnicities are vital for fostering innovation and holistic problem-solving. Each different ethnic background brings unique perspectives and experiences, enriching discussions and promoting deeper exploration of issues. Embracing this diversity cultivates inclusivity, understanding, and empathy among all cohort participants, leading to more robust and equitable solutions that can be overarching to climate emergency solutions.

<sup>11</sup> <https://www.bbc.com/news/science-environment-43294221> - Women are more likely than men to be affected by climate change, studies show. UN figures indicate that 80% of people displaced by climate change are women.



## 04 Diversity in entrepreneurship



**Josephine Bromley,**  
with co-founders **Ranjeet Bhalerao**  
and **Ruggiero Guida** from **Cohort 3**

### **Programmatic Content on Equity, Diversity, and Inclusion**

At Carbon13, Equity, Diversity, and Inclusion (EDI) isn't just a checkbox to tick—it's ingrained in every facet of our activities. From the initial selection of cohort participants to the ongoing support provided to invested companies by our experts, EDI principles guide our decision-making process. We prioritise inclusivity in our interactions, ensuring that voices from diverse backgrounds are not only heard but actively sought out and valued.

Throughout our programmes, we incorporate EDI-focused workshops, training sessions, and resources to ensure that cohort participants and, consequently, invested companies not only understand the importance of EDI but also actively work towards its realisation. By embedding EDI into the fabric of our programme, we empower our participants to become agents of change since the embryonic phase of their ventures, fostering inclusive cultures and driving meaningful impact within their future climatetech endeavours. By promoting spaces where everyone feels welcome and respected, we also drive innovation and growth through the diverse perspectives and experiences embraced.

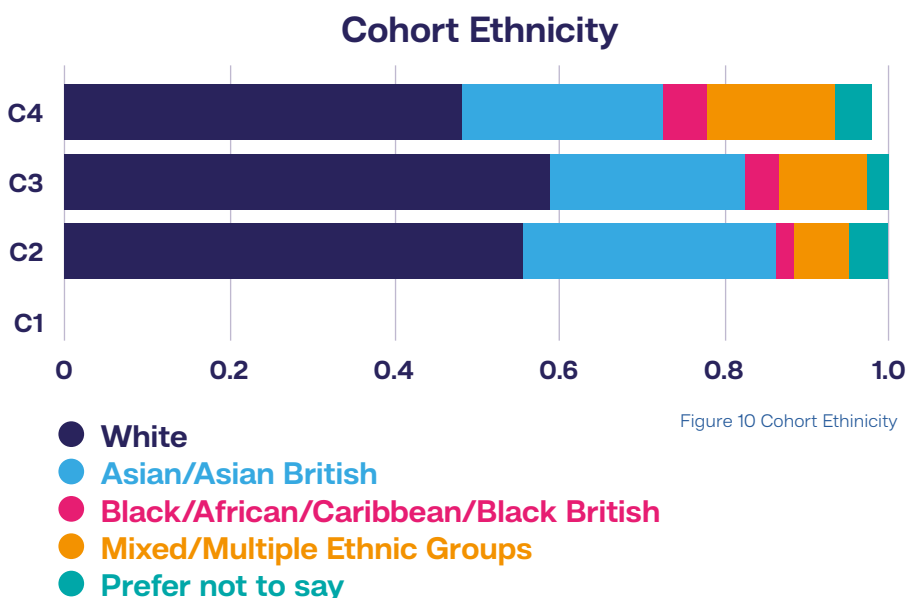


Figure 10 Cohort Ethnicity

# 05

## Conclusion

In 2021 and 2022, we are proud to say that we invested in 32 ventures that are inherently committed to solving the climate emergency, and created ground for many more to emerge.

Our startups and their bold teams are already making huge strides to decarbonise our world by changing car battery supply chains, helping save forests that otherwise would be destroyed, contributing to minimising emissions from fertilisers, revolutionising how cement is produced or how our water systems are managed. It is entrepreneurs, with their innovative spirit and drive, who are uniquely positioned to make a significant impact at scale and speed in these and in many other areas.

However, they can't go very far without the support from a whole ecosystem. A phrase we often use and believe in deeply is: real transformation doesn't come from a single individual but unencumbered communities with original thoughts. Throughout our initial years, we introduced more than 300 founders who bravely took action and faced a collective challenge. We would like to thank all of our stakeholders for trusting us to find great people and introduce a catalytic effect through our programmes to accelerate their missions to introduce innovation and effect change.

As stated previously, this is our first Impact Report, and we would like to acknowledge that reporting and providing transparency is an ongoing path. As Carbon13 grows and expands, and our ventures grow with us, we strive to gather more data and include more information regarding their impact.

We look forward to updating you further as our portfolio grows, our ventures scale, and we deliver more than one type of return.







# 06

## Methodology and Glossary

### Methodology

Assets and activities are assessed using carbon factors from government sources, reputable third party research, and xtonnes' proprietary modelling, with the most relevant factors being applied to your business. Where proxy data exists for how much an asset or activity has been used or carried out, this creates a "modelled" footprint based on the organisation in question, giving the most likely, lower and upper case possibilities.

Where data exists that measures the usage of an asset or completion of an activity, this creates a measure footprint, which typically receives an maximum quality score, and produces a single figure as opposed to a range.

### Glossary

**Additionality:** Additionality in climate-related matters refers to ensuring that claimed emission reductions are extra to what would naturally occur. This concept verifies that these claimed emission reductions wouldn't happen otherwise, emphasising their necessity in combating climate change.

**Greenhouse Gas:** Greenhouse gases (GHG) are gases that are responsible for the so-called greenhouse effect, which is considered the main cause of climate change. They can occur both naturally and anthropogenically (through man-made causes).

**Greenhouse Gas Protocol Corporate Accounting and Reporting Standards:** GHGP (Greenhouse Gas Protocol) Corporate Accounting and Reporting Standard is traditionally considered the "gold standard" in carbon accounting methodologies. Developed with the World Resources Institute (WRI) and World Business Council

for Sustainable Development (WBCSD), GHGP claims to provide the accounting platform for virtually every corporate GHG reporting program in the world.

**Carbon Footprint:** The total of the greenhouse gas emissions (both direct and indirect) associated with an organisation, product or activity.

**Carbon Dioxide Emissions (CO<sub>2</sub>):** Carbon dioxide emissions or CO<sub>2</sub> emissions are emissions stemming from the burning of fossil fuels and the manufacture of cement; they include carbon dioxide produced during consumption of solid, liquid, and gas fuels as well as gas flaring.

**Carbon Dioxide Equivalent (CO<sub>2</sub>e):** Carbon dioxide equivalent, or CO<sub>2</sub>e, is a metric measure representing all greenhouse gases by converting them to the equivalent amount of CO<sub>2</sub>. There are six greenhouse gases which contribute to the greenhouse effect, and thus to global warming,

according to the Kyoto Protocol: carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), nitrous oxide (laughing gas, N<sub>2</sub>O), hydrofluorocarbons (HFC/HFC), perfluorocarbons (PFC/PFC) and sulfur hexafluoride /SF<sub>6</sub>). Each has a different warming effect, so CO<sub>2</sub>e compares this with the effect of one unit of CO<sub>2</sub>.

**Scope 1 Emissions:** Scope 1 emissions are direct greenhouse gas emissions occurring from the combustion of fuels owned or controlled by the reporting organisation. They are also called direct emissions.

### There are three recognised sources of Scope 1 emissions:

- The burning of fossil fuels in a building or facility, e.g. a gas boiler or incinerator. This is the buildings' heat and power submitted on the xtonnes platform.
- Refrigerants, or fugitive emissions, from cooling equipment such as fridges or air conditioners at installation, in use or in disposal.
- The burning of fossil fuels in mobile sources (on-road and non-road vehicles or mobile machinery).

### Scope 2 Emissions:

Scope 2 emissions are the greenhouse gas emissions that result from a reporting organisation's purchased energy, usually electricity, but also heating, cooling, and steam. It includes the emissions associated with the transport and distribution of this energy to a reporting organisation. They are also called indirect emissions.

**Scope 3 Emissions:** Scope 3 emissions are "indirect" emissions associated with a business's upstream and downstream supply chains. They are the remainder of greenhouse gas emissions that the reporting organisation must have to carry out its activities. The Greenhouse Gas Protocol has 15 categories of Scope 3 emissions.

**Baseline Year:** A baseline year is the year from which emission reduction targets are set, or benchmarked. The percentage of the total emissions in the base year sets future reduction targets. To meet Science Based Targets, the baseline year should be 2018, or for small businesses, the closest year for which there is good data.

# 07

## Appendix

|                   |                     | 2022 |              |       | 2021 |              |       |
|-------------------|---------------------|------|--------------|-------|------|--------------|-------|
|                   |                     | Low  | Middle       | High  | Low  | Middle       | High  |
| Scope 1 & 2       | Heat                | 0.26 | <b>0.58</b>  | 0.69  | 2.65 | <b>5.77</b>  | 6.92  |
|                   | Power               | 0.23 | <b>0.28</b>  | 0.40  | 2.27 | <b>2.83</b>  | 3.99  |
|                   | Fugitives           |      |              |       | 0.08 | <b>6.50</b>  | 12.9  |
| Scope 1 & 2 TOTAL |                     | 0.49 | <b>0.86</b>  | 1.09  | 5.00 | <b>15.10</b> | 23.81 |
| Scope 3           | Wastes              | 0.03 | <b>1.76</b>  | 6.21  | 0.02 | <b>1.29</b>  | 4.55  |
|                   | Purchases           | 1.50 | <b>3.89</b>  | 6.28  | 0.29 | <b>0.60</b>  | 0.91  |
|                   | Waste Water         | 0.18 | <b>0.20</b>  | 0.23  | 0.13 | <b>0.15</b>  | 0.17  |
|                   | Trips               | 0.93 | <b>4.65</b>  | 5.92  | 0.64 | <b>0.80</b>  | 2.83  |
|                   | Commutes            | 0.26 | <b>0.48</b>  | 0.76  | 0.38 | <b>0.70</b>  | 1.12  |
|                   | Investments         | 3.03 | <b>3.03</b>  | 3.03  |      |              |       |
|                   | Remotes (non- GHGP) | 3.60 | <b>6.51</b>  | 9.61  | 2.64 | 4.78         | 7.05  |
|                   | Scope 3 TOTAL       | 9.53 | <b>20.52</b> | 32.05 | 4.10 | <b>8.31</b>  | 16.63 |
| COMBINED TOTAL    |                     |      | <b>21.38</b> |       |      | <b>23.41</b> |       |





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**Carbon13**

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